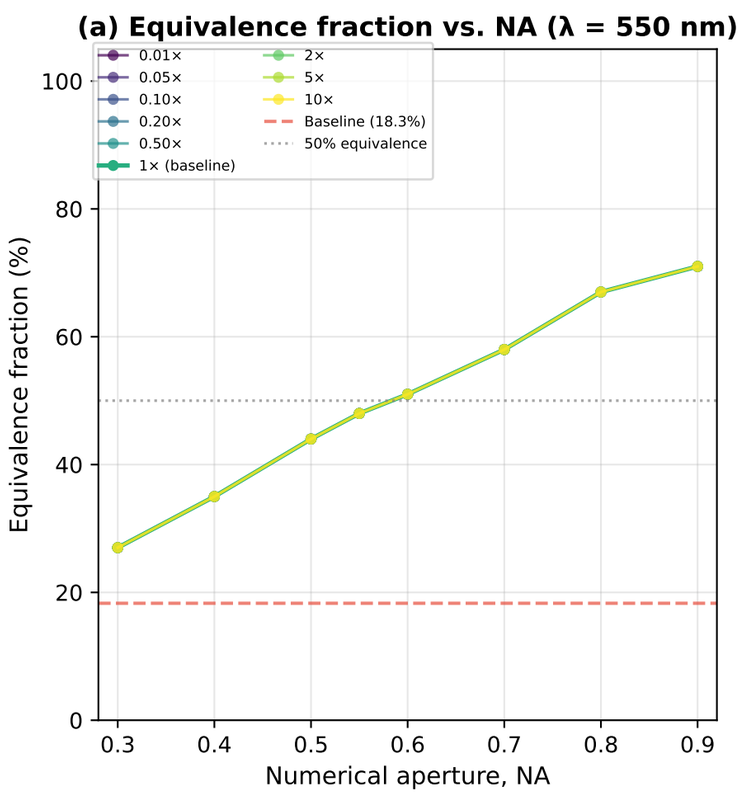

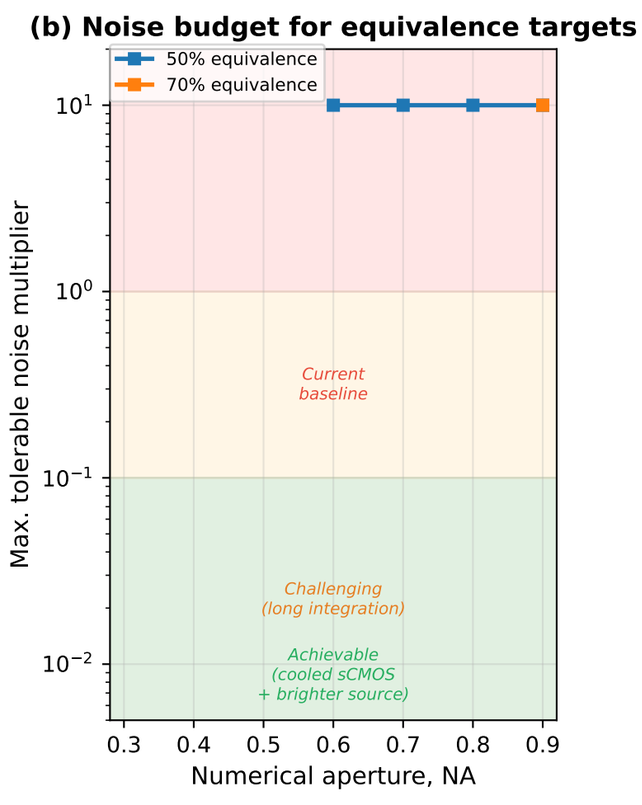

(c) Four improvements to close the gap

1. Reduce optical noise floor **100 ×**
Cooled sCMOS, brighter LED/laser, multi-frame averaging ($\sqrt{N}$ gain)
→ Single most impactful change

2. Increase NA to **0.7–0.9**
High-NA Mirau objectives (NA 0.8)
→ f_c from 1000 to 1450 mm^{-1}

3. Shorter **λ = 400–450** nm
Blue LED / diode laser
→ f_c reaches 2000 mm^{-1}

4. 2D phase unwrapping
Quality-guided or temporal PSI
→ 1% → 0.1% error rate

18% → 87% equivalence -- **13%** residual: physics-limited